

The Laplace transform at time zero.

1 Questions

When applying the inverse Laplace transform, sometimes there is an explicit unit step function ($\mu(t)$) included in the solution and sometimes it is omitted. E.g., when taking the inverse Laplace transform of $F(s) = 1/s$, I see the solution written as either $f(t) = \mu(t)$ or $f(t) = 1.0$.

Or another example: Given

$$f(t) = 10.0 (e^{-0.5t} \sin(2\pi t)),$$

we see the solution written as

$$F(s) = 10.0 \frac{2\pi}{(s + 0.5)^2 + (2\pi)^2}$$

or

$$F(s) = 10.0 \frac{2\pi}{(s + 0.5)^2 + (2\pi)^2} \mu(t)$$

Why is this the case?

2 Answer

The definitions of the Laplace transform and the inverse Laplace transform are key to understanding the details of functions at, and before $t = 0$,

$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t) e^{-st} dt \quad (1)$$

$$\mathcal{L}^{-1}[F(s)] = f(t)\mu(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds \quad (2)$$

where $\mu(t)$ is the unit step function defined as

$$\mu(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}. \quad (3)$$

Using the convention for the forward transform (2), we can see that the value of the function $f(t)$ prior to $t = 0^-$ not included within the bounds of integration¹ However, the inverse transform (??) requires that we define the function piecewise, with the value prior to time zero being defined as zero. This piecewise definition is realized by multiplication by $\mu(t)$.

If we followed these definitions strictly, it would be more correct to have independent tables for the forward Laplace transform pairs (Table 1) and inverse Laplace transforms pairs (Table 2). Because *all* of our functions are *always* equal to zero for $t < 0$, the multiplication by $\mu(t)$ will often be implicit, and we won't actually show $\mu(t)$, except in cases where it is needed for clarity².

¹See K. H. Lundberg, H. R. Miller, and D. L. Trumper, "Initial conditions, generalized functions, and the Laplace transform: Troubles at the origin," IEEE Control Systems, vol. 27, pp. 22-35, 2007. for a discussion of the important details on how the left bound of the transform is defined.

²Paraphrased from <https://lpsa.swarthmore.edu/LaplaceXform/FwdLaplace/LaplaceFuncs.html>

	$f(t)$	$F(s)$
1	$\delta(t)$ - Impulse	1
2	$\mu(t)$ - Step	$\frac{1}{s}$
3	t^n ($n = 1, 2, 3, \dots$)	$\frac{n!}{s^{n+1}}$
4	e^{-at}	$\frac{1}{s+a}$
5	$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
6	$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$

Table 1: Forward Laplace Transform Pairs

	$F(s)$	$f(t)\mu(t)$
1	1	$\delta(t)$ - Impulse
2	$\frac{1}{s}$	1 $\mu(t)$ - Step
3	$\frac{n!}{s^{n+1}}$	$t^n \mu(t)$ ($n = 1, 2, 3, \dots$)
4	$\frac{1}{s+a}$	$e^{-at} \mu(t)$
5	$\frac{\omega}{s^2 + \omega^2}$	$\sin(\omega t) \mu(t)$
6	$\frac{s}{s^2 + \omega^2}$	$\cos(\omega t) \mu(t)$

Table 2: Inverse Laplace Transform Pairs